Mode of Examination: Online

M.Tech. Semester – I Mid Sem Examination, 2021

2021

Subject: Computer Science

Paper Code & Name: CSCL 0901 Topics in Algorithms

Full Marks: 30

Date: 04.01.2022 **Time and Duration: 12.30 pm to 2.00 pm**

Please note the following instructions carefully:

- **Promise not to commit any academic dishonesty.**
- **Marks will be deducted if the same/similar answers are found in different answer-scripts.**
- **Candidates are required to answer in their own words as far as applicable.**
- **Each page of the answer scripts should have your University Roll # on the right-top corner. The name of the scanned copy of the answer script will be of the following format: Paper code-paper name-exam roll number.pdf (Example: CSCL0901-ToA-Midsem-97-CSM-201001.pdf)**
- **The subject of the mail should be the file name only.**
- **The name of the scanned answer-script is to be sent to cucsemtech@gmail.com**
- **Extra 10 minutes will be allowed for uploading the answer-script**
- **The answer-script may not be accepted after the scheduled time.**

Question 1 is compulsory. Answer any 4 questions from Q2-Q6

(1) Answer any 5 questions from the following: [2 marks x 5]

a) In randomized *select* algorithm for n elements, what is the probability that a worst case pivot is chosen in every call for select ?

b) Let us consider a situation when inputs are nearly sorted. Upon calling quicksort on a subarray with fewer than k elements, let it simply return without sorting the subarray. After the top-level call to quicksort returns, we run insertion sort on the entire array to finish the sorting process. Argue that this sorting algorithm runs in $O(nk + n \log n/k))$ expected time.

c) Show the decision tree to sort three elements a, b, c.

d) Write the recurrence equation and solve it if select algorithm chooses a group size of 11.

e) In randomized closest pair algorithm why do you check distances between only the points in 25 neighboring grid cells of the current point that is processed ?

f) Show that the probability of two keys having same hash value is only $1/n$ when coefficients of universal hash function are chosen uniformly at random.

(2) Suppose you are given an array $A[1..n]$ such that the values in array increases upto index position p and then decreases till end. Write a $O(\log n)$ algorithm to find the position p. Justify its time complexity [5].

(3) Give the node structure of a skip list. Write delete function to delete a given key from the skip list. Assume there is a pointer ptr pointing to the node to be deleted after performing a $search$ operation in the skip list. What is the time complexity? Justify. [5]

(4) Let g denote the size of each group in divide and conquer $Select$ algorithm that find k^{th} smallest element. Derive the running time of the algorithm in terms of g. What happens when g is too large compared to 5? [5]

(5) Modify the divide and conquer *closest pair* algorithm such that the Y array is not pre sorted yet the running time remains $O(n \log n)$. [5]

(6) Explain how the hash table is used in randomized closest pair. Prove that the expected number of insert operations in the Hash Table in the randomized closed pair algorithm is $O(n)$. [5]